

भारतीय मानक
Indian Standard

IS 1417 : 2016

**स्वर्ण एवं स्वर्ण मिश्रधातुएँ,
आभूषण/शिल्पकारी शुद्धता एवं
मुहरांकन — विशिष्टि**
(चौथा पुनरीक्षण)

**Gold and Gold Alloys, Jewellery/
Artefacts — Fineness and
Marking — Specification**
(*Fourth Revision*)

ICS 39.060

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भारतीय मानक ब्यूरो
BUREAU OF INDIAN STANDARDS
मानक भवन, 9 बहादुरशाह ज़फर मार्ग, नई दिल्ली-110002
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI-110002
www.bis.org.in www.standardsbis.in

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FOREWORD

This Indian Standard (Fourth Revision) was adopted by Bureau of Indian Standards, after the draft finalized by the Precious Metals Sectional Committee had been approved by the Metallurgical Engineering Division Council.

This standard was first published in 1959 and subsequently revised in 1971, 1981 and 1999. In many advanced countries, the gold ornaments/artefacts traded are certified for purity and are marked with various details like purity/fineness of the ornament, manufacturer's mark, the mark of the assaying laboratory responsible for certifying the purity, etc. These ornaments/artefacts are termed as Hallmarked.

This standard is primarily meant to classify the various grades of gold according to their gold content and provide guidelines to mark the gold articles. The percentage of alloying elements stipulated for various grades depend upon their specific uses and the detailed compositions of these grades have been covered in IS 2790 : 1999 'Guidelines for manufacture of 23, 22, 21, 18, 14 and 9 carat gold alloys (*second revision*)'.

In this revision, bullion and coin have been defined and included in the scope of the standard and marking clause has been suitably modified, the grades of fineness 958(23k), 875(21k), 833 (20k), 791(19k), 708(17k), 666(16k) and 375(9k) have been deleted in consumer welfare, and a negative tolerance of maximum 2ppt has been introduced on re-testing of hallmarked jewellery/artefact.

Assistance has been derived from International Standard ISO 9202 : 2014 'Jewellery — Fineness of precious metal alloys', issued by the International Organization for Standardization.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

GOLD AND GOLD ALLOYS, JEWELLERY/ARTEFACTS
— FINENESS AND MARKING — SPECIFICATION
(Fourth Revision)

1 SCOPE

1.1 This standard specifies based on gold content:

- a) Grades of fine gold and standard gold used for manufacture of bullion and coins, and
- b) Grades of gold alloy used in manufacture of jewellery/artefacts.

1.2 This standard also specifies the guidelines for marking of purity and other details on tested bullion, coins and jewellery/artefacts.

NOTE — Artefacts cover medallions and utensils.

2 REFERENCES

The following standards contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standard indicated below:

<i>IS No.</i>	<i>Title</i>
1418 : 2009	Assaying of gold in gold bullion, gold alloys and gold jewellery/artefacts — Cupellation (fire assay) method (<i>second revision</i>)
3095 : 1999	Gold solders for use in manufacture of jewelleryd Specification (<i>second revision</i>)
15820 : 2009	General requirements for component or assaying and hallmarking centre

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Assaying — The method of accurate determination of the gold content of the sample expressed in parts per thousand (‰).

3.2 Carat — Theratio between the mass of gold content and the total mass expressed in parts per twenty four.

3.3 Fineness — The ratio between the mass of gold content and the total mass expressed in parts per thousand (‰).

3.4 Fine Gold — It is gold having fineness 999 parts per thousand (‰) and above.

3.5 Gold — The metallic element gold free from any other element.

3.6 Gold Alloy — Gold alloyed with one or more other elements.

3.7 Standard Gold — Gold having fineness 995 parts per thousand (‰) and above.

3.8 Bullion — Refined gold bars of various shapes of standard and fine gold manufactured by mints/refineries.

3.9 Coin — Gold coin of standard and fine gold manufactured by mints /refineries.

4 GRADES

4.1 Gold and gold alloys shall be classified in accordance with the following grades depending upon their fineness:

<i>Grade</i>	<i>Fineness in ppt Min</i>	<i>Purity in Carat (K)</i>
(1)	(2)	(3)
Fine gold	999	—
Standard gold	995	—
916.6	916	22
750	750	18
585	585	14

4.1.1 The above grades shall fulfil the following:

- a) The above classification is applicable for gold jewellery/artefacts also.
- b) The maximum permissible limits of Cadmium and each of Platinum group metals in gold alloys for manufacturing jewellery/artefacts and gold solders are 0.02 percent and 0.05 percent respectively.
- c) Presence of ruthenium, iridium and cadmium in gold and gold alloys shall be detected by X-ray fluorescence (XRF) method.

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5 REQUIREMENTS

5.1 Solders used for the manufacture of gold jewellery/artefacts shall be of the same fineness as that of the jewellery/artefacts and shall confirm to the requirement as specified in IS 3095. Solder compositions other than specified in IS 3095 may also be used provided their gold content is same as the respective grade as specified for the standard solder and their Cadmium content is within permissible limit as given in **4.1.1 (b)**.

5.2 Sampling

Unless specified otherwise for the jewellery/artefact of the same design and quality, the sample should be taken from 10 percent of the jewellery/artefacts selected at random (*see* IS 15820). In all other cases, the sample should be taken from each jewellery/artefact. Any other sampling plan may also be followed subject to agreement between the contracting parties or as specified by the inspection agency as the case may be.

5.2.1 The sampling of bullion and coins shall be done as specified in IS 1418/inspection agency.

5.3 Assaying

The sample taken as prescribed in **5.2** shall be assayed in accordance with IS 1418. The actual weight of samples, etc shall however be as prescribed in respective test method detailed in IS 1418.

5.4 Tolerances

No negative tolerance shall be allowed on the gold content of gold jewellery/artefact, bullion, and coinsampled and tested as per **5.2** and **5.3** respectively.

5.5 Re-testing (Testing of Hallmarked Jewellery/Artefacts)

In case of testing of the hallmarked jewellery/artefact, a negative tolerance of maximum 2 ppt is permitted.

5.6 Conformity

For ascertaining conformity of a sample to the declared fineness, the mean value of the duplicate determinations as per IS 1418 shall be used

6 MARKING

6.1 Hallmarking shall be done in either of the following manner:

- a) The gold, gold alloys, jewellery/artefacts shall be hallmarked by BIS recognized assaying and hallmarking centres only. The

hallmarked articles shall carry the following makings:

- 1) BIS Standard Mark;
 - 2) Purity in caratand fineness;
 - 3) Assay centre's identification mark/number; and
 - 4) Jeweller's identification mark/number.
- b) Hallmarked gold bullion and gold coins shall have the following markings:
- 1) BIS Standard Mark;
 - 2) Fineness;
 - 3) Name/Identification of the manufacturer; and
 - 4) Serial Number

NOTE — In case of coins, serial numbers may be marked on the packaging.

6.1.1 Hallmarking

The use of Hallmark is governed by the provisions of the *Bureau of Indian Standards Act, 1986* and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of the Standard Mark may be granted to jewellers/jewellery manufacturers may be obtained from the Bureau of Indian Standards.

6.1.2 Marking shall be legible and durable.

6.1.3 Marking shall be done on all parts which can be easily removed or replaced.

6.2 Goldjewellery/artefacts on which hallmarking shall not be applied are as follows:

- a) Any gold jewellery/artefact which has not been assayed by BIS recognized Assaying and Hallmarking centres.
- b) Any goldjewellery/artefactwhich fails to comply with declared fineness and caratage.
- c) Any gold jewellery/artefact which is not solid or which is made with a hollow centre and then filled with base metal, cement, lac or other foreign substance thereby giving the article a false or added weight or strength unless the weight of gold is declared by the jeweller and marked on the article by the centre.
- d) Gold jewellery/artefacts on which it is physically impossible to stamp the marks.

6.2.1 Gold jewellery/artefact of weight less than 2 g may be hallmarked.

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Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002

Telephones : 2323 0131, 2323 3375, 2323 9402

Website: www.bis.org.in

Regional Offices:

Telephones

Central : Manak Bhavan, 9 Bahadur Shah Zafar Marg
NEW DELHI 110002

{ 2323 7617
2323 3841

Eastern : 1/14 C.I.T. Scheme VII M, V. I. P. Road, Kankurgachi
KOLKATA 700054

{ 2337 8499, 2337 8561
2337 8626, 2337 9120

Northern : SCO 335-336, Sector 34-A, CHANDIGARH 160022

{ 260 3843
260 9285

Southern : C.I.T. Campus, IV Cross Road, CHENNAI 600113

{ 2254 1216, 2254 1442
2254 2519, 2254 2315

Western : Manakalaya, E9 MIDC, Marol, Andheri (East)
MUMBAI 400093

{ 2832 9295, 2832 7858
2832 7891, 2832 7892

Branches: AHMEDABAD. BENGALURU. BHOPAL. BHUBANESHWAR. COIMBATORE. DEHRADUN.
FARIDABAD. GHAZIABAD. GUWAHATI. HYDERABAD. JAIPUR. KOCHI. LUCKNOW.
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